The Wave That Should Not Exist

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1 Introduction

It is easy to observe a wave in water—almost everyone has thrown a stone into a lake or river and watched the circular ripples spread outward and weaken quickly. This is the normal experience most of us have with waves. If the energy of waves on an ocean beach is sustained, it is due to the wind we can feel. What about sitting on a beach and observing a single wave without visible force? Our experience says that it should not be possible. Yet such a wave, called a *soliton*, does exist.

The best known first record of a soliton in a mathematical journal was made by the Scottish engineer John Scott Russell:[1]

> "I was observing the motion of a boat which was rapidly drawn along a narrow channel by a pair of horses, when the boat suddenly stopped—not so the mass of water in the channel which it had put in motion; it accumulated round the prow of the vessel in a state of violent agitation, then suddenly leaving it behind, rolled forward with great velocity, assuming the form of a large solitary elevation, a rounded, smooth and well-defined heap of water, which continued its course along the channel apparently without change of form or diminution of speed. . . . I followed it on horseback, and overtook it still rolling on at a rate of some eight or nine miles an hour, preserving its original figure some thirty feet long and a foot to a foot and a half in height. . . . Such, in the month of August 1834, was my first chance interview with that singular and beautiful phenomenon which I have called the Wave of Translation."

Indeed, a wave traveling for such a long time with little change is remarkable; see Figure 1 for the commemorative plaque. Assuming it took John several minutes to get on the horse, ride, and catch up to the wave, it covered roughly a mile and a half at 9 mph for guesstimated about 10 minutes. The wave was travelling in the same shape, surviving interaction with canal banks.

2 Describing and Solving a Soliton Mathematically

Common water waves break down due to dispersion and nonlinearity. Dispersion means that different parts of the wave move at different speeds thus the wave flattens and stretches.

[1] John Scott Russell, "Report on Waves," in *Report of the Fourteenth Meeting of the British Association for the Advancement of Science* (London: John Murray, 1845), 311–390.

Figure 1: Commemorative plaque reads "Near this spot JOHN SCOTT RUSSELL discovered the SOLITARY WAVE in August 1834."



Source: Graeme, "Discovering Mathematical Tourism," February 25, 2010, `https://maths.straylight.co.uk/archives/256`.

Nonlinearity means that the wave steepens and distorts, as the taller parts of the wave move faster, shorter parts lag behind, the wave steepens, forming shocks or breaking. In a soliton wave, dispersion perfectly cancels nonlinearity, therefore the wave keeps its form, moves at a constant speed, and survives collisions [2] or interactions with canal walls or banks.

In 1895, the Dutch mathematician Diederik Korteweg, a professor of mathematics, mechanics, and astronomy at the University of Amsterdam, together with his doctoral student Gustav de Vries, derived a partial differential equation whose solutions include solitary waves.

The logic was as follows. We consider the simplest case in only one space variable x and look for a function u (height, or deviation from equilibrium), depending on the variables x and t, i.e. $u(x,t)$ describes the one-dimensional wave's height at the place x at time t. Using the typical short notations [3] ,

$$
\begin{aligned}
u_t(x,t) &= \frac{\partial u(x,t)}{\partial t} && \text{(speed of height change at one location)},\\
u_x(x,t) &= \frac{\partial u(x,t)}{\partial x} && \text{(slope of the wave)},\\
u_{xx}(x,t) &= \frac{\partial^2 u(x,t)}{\partial x^2} && \text{(curvature)},\\
u_{xxx}(x,t) &= \frac{\partial^3 u(x,t)}{\partial x^3} && \text{(change of curvature)}.
\end{aligned}
$$

The resulting equation, now known as the Korteweg–de Vries (KdV) equation, can be

[2] Even in a collision of two solitons, each of them emerges with the same shape and speed. Such behavior is more typical of particles than waves.

[3] Klaus Brauer (University of Osnabrück, Department of Mathematics & Computer Science), "The Korteweg-de Vries Equation: History, exact Solutions, and graphical Representation," May 2000, accessed January 2026, `https://math.arizona.edu/~gabitov/teaching/141/math_485/KDV.pdf`.

written as

$$u_t(x,t) + 6\,u(x,t)\,u_x(x,t) + u_{xxx}(x,t) = 0. \tag{1}$$

In this equation, term $6uu_x$ is responsible for the fact that higher parts of the wave move faster, which steepens the wave; mathematically, value of the function affects its rate of change with respect to time represented by the first term, u_t. The term u_{xxx} represents dispersion which acts to spread the wave out. Most often, either the force of steepening or the force of dispersion dominates. The wave tends to break if steepening dominates, or fades if dispersion dominates. The KdV equation captures the mathematically perfect balance between the two forces, cancelling them exactly, and allowing the wave to become a solitary one.

A note on the coefficient 6 in the term uu_x. Mathematically, the coefficient 6 is introduced without loss of generality, meaning that with another coefficient derived from related physics constants the equation $u_t + auu_x + u_{xxx} = 0$ would describe the same physics, on a less convenient scale mathematically, and the *soliton* solution will still exist. The KdV's coefficient of 6 is a convenient rescaling of variables that simplifies the mathematics, making it elegant.

Since the wave keeps its form while travelling, this suggests seeking solutions in the form $u(x,t) = f(x-ct)$ where c is the constant speed of travel to the right of the whole profile defined by the same f. Based on this representation, we will convert the equation into single-variable calculus denoting substitution $y = x - ct$ (further, f' is understood as a derivative with respect to y):

$$u_t(x,t) = \frac{\partial f(x-ct)}{\partial t} = -cf'(y), u_x = f'(y), \text{ and } u_{xxx} = f'''(y)$$

Then KdV equation becomes $-cf' + 6ff' + f''' = 0$, and we can integrate both sides, collecting constants on the right (denoted A):

$$-cf + 3f^2 + f'' = A$$

We note that $f(y) \to 0, \quad f'(y) \to 0$, and $\quad f''(y) \to 0 \quad$ as $|y| \to \infty$, forcing $A = 0$ and the equation to become $f'' = cf - 3f^2$.

Omitting certain justifications, we multiply both sides by f' and then integrate both sides, denoting the constant B:

$$f''f' = cff' - 3f^2f'$$

$$\frac{1}{2}(f')^2 = \frac{c}{2}f^2 - f^3 + B$$

Again, since $f(y) \to 0$ and $\quad f'(y) \to 0 \quad$ as $|y| \to \infty$, it forces $B = 0$. The equation becomes

$$(f')^2 = cf^2 - 2f^3 = f^2(c - 2f)$$

This equation is known from physics because it is similar to a conservation-of-energy equation with the left side representing kinetic energy and the right side representing potential energy. It follows that $0 \le f \le \frac{c}{2}$, the right inequality due to the fact that permitting

f above $\frac{c}{2}$ would make the equation right-hand side negative while the left-hand side is a square.

tThe solution to such equation is also known and it is hyperbolic secant sech (which is the reciprocal of hyperbolic cosine cosh) $sech(z) = \dfrac{2}{e^z + e^{-z}}$. The equation solution is:

$$f(y) = \frac{c}{2} sech^2\left(\frac{\sqrt{c}}{2} y\right) \text{ and } u(x,t) = \frac{c}{2} sech^2\left(\frac{\sqrt{c}}{2}(x - ct)\right).$$

3 Modern View

After almost 70 years of little progress since the re-discovery of the KdV equation, in 1965 the successful application of computer-based numerical methods by Norman Zabusky at Bell Labs and Martin Kruskal at Princeton University revived interest in the soliton phenomenon.[4] This work also pointed to hierarchies of solitons, structures in nonlinear soliton waves, and possible solutions to more complex mathematical models.

One of the best applications of soliton theory was found in transmitting signals via fiber-optic cables.[5] The ability of soliton pulses to travel along optical fiber while maintaining their original wave shape makes solitons an attractive choice for transmitting data signals over very long distances.

[4]N. J. Zabusky and M. D. Kruskal, "Interaction of 'Solitons' in a Collisionless Plasma and the Recurrence of Initial States," *Physical Review Letters* 15, no. 6 (August 9, 1965): 240–243, `https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.15.240`.

[5]Alex Kasman, "A Brief History of Solitons and the KdV Equation," *Current Science* 115, no. 8 (October 25, 2018): 1486–1496, `https://www.jstor.org/stable/10.2307/26978440`.